THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Cell theory was proposed by: 1

(a) Hooke and Brown **(b)** Watson and Crick
(c) Darwin and Wallace **(d)** Schleiden and Schwann

2. Muscles attached to bones and used in running and writing are: 1

(a) involuntary branched muscles **(b)** cardiac muscles
(c) smooth muscles **(d)** striated (voluntary) muscles

3. Fish breathe through: 1

(a) lungs **(b)** skin only **(c)** trachea **(d)** gills

4. Sperms are produced in the: 1

(a) ovaries **(b)** oviduct **(c)** testes **(d)** uterus

5. The continuous movement of water between the oceans, atmosphere and land is the: 1

(a) rock cycle **(b)** carbon cycle
(c) water (hydrological) cycle **(d)** nitrogen cycle

6. Which of the following is a prokaryote? 1

(a) yeast **(b)** amoeba **(c)** onion cell **(d)** bacterium

7. The tissue responsible for the growth of plants is the: 1

(a) nervous tissue **(b)** meristematic tissue
(c) permanent tissue **(d)** epithelial tissue

8. The science of classification of organisms is: 1

(a) ecology **(b)** taxonomy **(c)** genetics **(d)** physiology

9. Amoeba reproduces by: 1

(a) binary fission **(b)** fragmentation **(c)** budding **(d)** spore formation

10. The solid Earth including rocks, soil and the interior is the: 1

(a) geosphere **(b)** biosphere **(c)** atmosphere **(d)** hydrosphere

11. The smooth ER is concerned with: 1

(a) protein synthesis **(b)** lipid synthesis and detoxification
(c) respiration **(d)** photosynthesis

12. Water and minerals are conducted upward in plants by the: 1

(a) parenchyma **(b)** collenchyma **(c)** phloem **(d)** xylem

13. The 'amphibians of the plant kingdom' are the: 1

(a) bryophytes **(b)** algae **(c)** pteridophytes **(d)** gymnosperms

14. The female reproductive part of a flower is the: 1

(a) stamen **(b)** petal **(c)** sepal **(d)** carpel (pistil)

15. The surface that reflects the least solar radiation is: 1

(a) ice **(b)** snow
(c) ocean water or forests **(d)** clouds

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Name three types of plastids and their functions. 2

17. Differentiate between xylem and phloem in function. 2

18. State two features of the kingdom Fungi. 2

19. How does bread mould reproduce? 2

20. What is a biogeochemical cycle? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe the characteristics of Phylum Porifera and Cnidaria. 3

22. Name the features of insect-pollinated and wind-pollinated flowers. 3

23. Explain why the Earth's temperature at the equator and poles differ with the angle of sunlight. 3

24. State three functions of vacuoles. 3

OR

Differentiate between voluntary and involuntary muscles with examples. 3

25. Name the group (phylum or class) to which each belongs: 3

(a) Hydra
(b) tapeworm
(c) cockroach
(d) frog
(e) bat.

OR

State three methods of contraception with an example each. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the Earth as a system, describing the four spheres, the flow of energy and the cycling of matter. 5

OR

Describe the two types of cell division and their significance. 5

27. Describe the muscular tissue with the three types, structure and function of each in a table of five points. 5

OR

Compare monocots and dicots in a table of five points with examples. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND